AI-BASED EYE DISEASE IDENTIFICATION & HEALTH RECOMMENDATION SYSTEM

BATCH NO: 33 | 23551A42B3 • 23551A4275 • 24555A4207

CLINICAL OPHTHALMOLOGY & EXPLAINABLE AI DIAGNOSTIC REPORT

Patient Name: Ramesh Patel	**Patient ID:** PAT-2609-8FF2	**Exam Date:** 23 Sep 2026, 07:12 AM
Age / Gender: 58 yrs / Male	**Contact:** None	**Medical History:** Type-2 Diabetes Mellitus (12 years), Mild Hypertension. HbA1c: 8.4%

1. ARTIFICIAL INTELLIGENCE DIAGNOSTIC SUMMARY

Detected Condition	Severity Staging	AI Confidence	Overall Risk Index
Age-Related Macular Degeneration (AMD)	**Severe**	91.0%	70.0 / 100

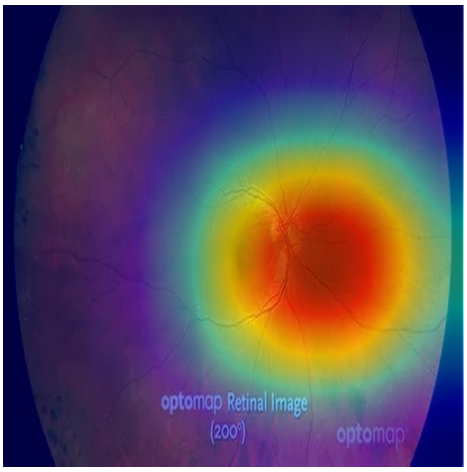
Cup-to-Disc Ratio (CDR): 0.6 **Macular Integrity:** 29.0% **Lesion Density:** 95.0% **Vessel Tortuosity:** 1.75

2. EXPLAINABLE AI (XAI) GRAD-CAM LESION LOCALIZATION



Original Retinal Fundus

Full vascular tree & optic disc



Grad-CAM Explainability Heatmap

Warm regions (Red/Yellow) highlight focal lesions

3. PERSONALIZED NUTRITIONAL & EYE CARE PRESCRIPTION

Target Nutrients:

• **AREDS-2 Formulation Elements**: Clinically proven to reduce progression risk from intermediate to advanced neovascular AMD. (*Food sources: Lutein (10mg), Zeaxanthin (2mg), Vit C, Vit E, Zinc, Copper*)

• **Zinc & Copper Balance**: Highly concentrated in choroid and retina, essential for retinal pigment epithelium metalloenzymes. (*Food sources: Pumpkin seeds, oysters, lentils, cashews*)

• **Anthocyanins**: Strengthens rod/cone photoreceptor regeneration and promotes dark adaptation. (*Food sources: Wild bilberries, blackberries, blackcurrants*)

Foods to Avoid:
• **High Glycemic Index Starches**: White bread, pastries, and processed potatoes correlate with faster AMD progression.
• **Hydrogenated Vegetable Oils**: Displaces essential retinal DHA in photoreceptor lipid bilayers.

Daily Eye Care Habits (20-20-20 Rule):
• Weekly self-monitoring with an Amsler Grid (monocularly) to catch waviness or dark spots early.
• Wear amber-tinted lenses to minimize blue-light induced photo-oxidative stress on RPE cells.

CLINICAL ACTION REQUIRED: High Severity - Immediate Specialist Referral

URGENT: Visit a specialized eye hospital or retina/glaucoma consultant within 24 to 72 hours.

System Certification:

AI-assisted screening tool for clinical decision support. Final diagnostic confirmation must be validated by a registered ophthalmologist.

Consulting Ophthalmologist:

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Signature & Stamp